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Local Discontinuity Measures for 3-D Seismic Data

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ABSTRACT

In this work, an analysis method is developed for the robust and efficient estimation of 3-D seismic local structural entropy, which is a measure of local discontinuity. This method avoids the computation of large covariance matrices and eigenvalues, associated with the eigenstructure-based and semblance-based coherency estimates. We introduce a number of local discontinuity measures, based on the relations between subvolumes (quadrants) of the analysis cube. The scale of the analysis is determined by the type of geological feature that is of interest to the interpreter. By combining local structural entropy volumes using various scales, we obtain a higher lateral resolution and better discrimination between incoherent and coherent seismic events. Furthermore, the method developed is computationally much more efficient than the eigenstructure-based coherency method. Its robustness is demonstrated by synthetic and real data examples.